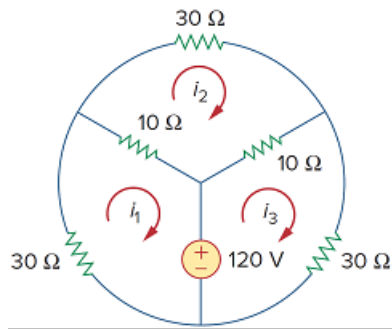


Problem

Find i_1 , i_2 , and i_3 in the circuit of Fig. 3.103.

Figure 3.103:



Step-by-step solution

Step 1 of 5

Refer Figure 3.103 in the textbook.

Let i_1, i_2, i_3 be the currents flowing through mesh 1, mesh 2 and mesh 3.

Apply Kirchhoff's voltage law to the loop 1.

$$10(i_1 - i_2) + 120 + 30i_1 = 0$$

$$40i_1 - 10i_2 + 120 = 0 \quad \dots (1)$$

Step 2 of 5

Apply Kirchhoff's voltage law to the loop 3.

$$-120 + 10(i_3 - i_2) + 30i_3 = 0$$

$$40i_3 - 10i_2 - 120 = 0 \quad \dots (2)$$

Apply Kirchhoff's voltage law to the loop 2.

$$10(i_2 - i_1) + 10(i_2 - i_3) + 30i_2 = 0$$

$$50i_2 - 10i_1 - 10i_3 = 0 \quad \dots (3)$$

Step 3 of 5

Subtract equation (2) from equation (1).

$$40i_1 - 10i_2 + 120 = 0$$

$$\underline{40i_3 - 10i_2 - 120 = 0}$$

$$40(i_1 - i_3) = -240$$

$$i_1 - i_3 = -6 \dots\dots (4)$$

Substitute equation (4) in equation (3)

$$50i_2 - 10i_1 - 10i_3 = 0$$

$$10(i_1 + i_3) = 50i_2$$

$$-6 + i_3 + i_3 = 5i_2$$

$$-3 + i_3 = 2.5i_2$$

$$i_3 = 2.5i_2 + 3 \dots\dots (5)$$

Step 4 of 5

Substitute $2.5i_2 + 3$ for i_3 in equation (2).

$$40(2.5i_2 + 3) - 10i_2 - 120 = 0$$

$$90i_2 = 0$$

$$i_2 = 0 \text{ A}$$

Therefore, the value of i_2 is 0 A.

Step 5 of 5

Substitute 0 A for i_2 in equation (5).

$$i_3 = 2.5(0) + 3$$

$$= 3$$

Therefore, the value of i_3 is 3 A.

Substitute 3 A for i_3 in equation (4).

$$i_1 = -6 + i_3$$

$$= -3$$

Therefore, the value of i_1 is -3 A.